WORKED SOLUTION

STEP I 1987, Question 13

Mechanics • Variable Forces

STEP I 1987, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q13

The key idea

The force law changes at $x = 0$, so the motion splits into stages, each governed by a simple linear differential equation. On the positive side the force is a restoring force proportional to displacement, which is simple harmonic motion; on the negative side it is a resistance proportional to velocity, which gives exponential decay of the speed. The position and velocity at the end of one stage are the initial conditions for the next. The real question is how many stages there are: does the particle ever come back?

Solution

Model

Treat the body as a particle on the x-axis acted on only by the given force, and write $\dot{x} = \mathrm{d}x/\mathrm{d}t$. Newton's second law, $m\ddot{x} = \mathrm{F}$, gives

$$\ddot{x} = -\mu^2 x \quad (x \geq 0), \qquad\qquad \ddot{x} = -\kappa\dot{x} \quad (x < 0).$$

The mass cancels, so it plays no part in the answer.

Stage 1: simple harmonic motion while $x \geq 0$

Since $v_0 > 0$, the particle immediately enters $x > 0$, where $\ddot{x} = -\mu^2 x$. This is simple harmonic motion with angular frequency μ, so its general solution is

$$x = A\cos\mu t + B\sin\mu t.$$

The condition $x = 0$ at $t = 0$ gives $A = 0$, and $\dot{x} = B\mu = v_0$ at $t = 0$ gives $B = v_0/\mu$. So

$$x = \frac{v_0}{\mu}\sin\mu t, \qquad \dot{x} = v_0\cos\mu t.$$

This stays valid while $x \geq 0$, that is while $\sin \mu t \geq 0$, so for $0 \leq t \leq \pi/\mu$. The particle reaches a greatest displacement v_0/μ at $t = \pi/(2\mu)$ and returns to the origin at

$$t = T = \frac{\pi}{\mu}, \qquad \text{with velocity} \qquad \dot{x} = v_0 \cos \pi = -v_0.$$

It comes back with the same speed it started with, as energy is conserved on this side (the force $-m\mu^2 x$ comes from the potential energy $\frac{1}{2}m\mu^2 x^2$, which is zero at the origin).

Stage 2: resisted motion while $x < 0$

Moving with velocity $-v_0$, the particle now enters $x < 0$, where $\ddot{x} = -\kappa \dot{x}$. Writing $u = \dot{x}$, this is the first-order equation $\mathrm{d}u/\mathrm{d}t = -\kappa u$, whose solution with $u = -v_0$ at $t = T$ is

$$\dot{x} = -v_0 \, \mathrm{e}^{-\kappa(t-T)}.$$

Integrating, with $x = 0$ at $t = T$:

$$x = \int_T^t -v_0 \, \mathrm{e}^{-\kappa(s-T)} \, \mathrm{d}s = -\frac{v_0}{\kappa} \left(1 - \mathrm{e}^{-\kappa(t-T)}\right).$$

Ask yourself Does the particle ever return to $x \geq 0$, starting a third stage?

No. The velocity $-v_0 \mathrm{e}^{-\kappa(t-T)}$ is negative for every $t > T$, so x decreases throughout this stage; it is strictly negative for all $t > T$ and the formula above holds for ever. The resistance slows the particle but never reverses it. As $t \to \infty$, $\mathrm{e}^{-\kappa(t-T)} \to 0$, so the particle approaches the point $x = -v_0/\kappa$ without ever reaching it.

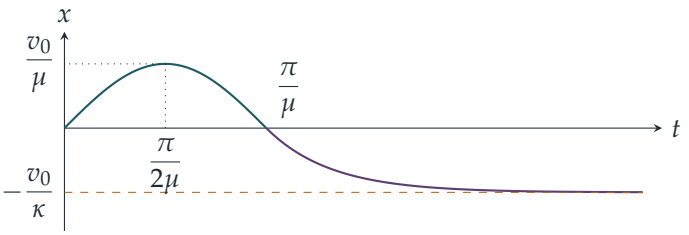
Answer

$$x = \begin{cases} \dfrac{v_0}{\mu} \sin \mu t, & 0 \leq t \leq \dfrac{\pi}{\mu}, \\ -\dfrac{v_0}{\kappa} \left(1 - \mathrm{e}^{-\kappa(t-\pi/\mu)}\right), & t > \dfrac{\pi}{\mu}. \end{cases}$$

Both pieces have $x = 0$ and $\dot{x} = -v_0$ at $t = \pi/\mu$, so the graph is smooth there (with a continuous gradient), although $\ddot{x}$ jumps from 0 to κv_0.

Sketch

The graph is half an arch of a sine curve above the axis, followed by a curve that leaves the origin with the same slope $-v_0$ and levels off towards the asymptote $x = -v_0/\kappa$. (The sketch uses $\mu = 1$, $\kappa = 1$.)



In general the peak height v_0/μ and the depth of the asymptote v_0/κ are independent: a weak resistance (small κ) lets the particle drift a long way to the left.

Checks

- Units: μ and κ both have units of s^{-1}, so v_0/μ and v_0/κ are lengths, as they must be.
- Limiting case: as $\kappa \to 0$ there is no resistance, and indeed $-\frac{v_0}{\kappa}(1 - \mathrm{e}^{-\kappa s}) \to -v_0 s$, uniform motion at speed v_0.
- Numerically: stepping the equation of motion with a Runge–Kutta method for $\mu = 1.7$, $\kappa = 0.9$, $v_0 = 2.2$ matched the formula to within 3×10^{-5} for $0 \le t \le 15$; the particle peaked at $x = 1.2941 = v_0/\mu$ and settled at $x = -2.4444 = -v_0/\kappa$.

Common slips

- Measuring time in the second stage from $t = 0$ rather than from $t = \pi/\mu$: the exponent must be $-\kappa(t - \pi/\mu)$.
- Forgetting to check whether the particle re-enters $x \ge 0$; the question asks for the *whole* subsequent motion, so the claim that there is no third stage needs its one-line justification.

Beyond STEP

Here the force is a different linear expression on each side of $x = 0$, a piecewise-linear system; the trick of gluing solutions together, matching position and velocity at the switching point, is the standard way such systems (and impact and friction problems) are handled in a first course in differential equations. Adding both terms at once, $\ddot{x} + \kappa\dot{x} + \mu^2 x = 0$, gives the damped harmonic oscillator, whose underdamped, critically damped and overdamped regimes are a central example in any first mechanics course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the damped oscillator carefully.

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